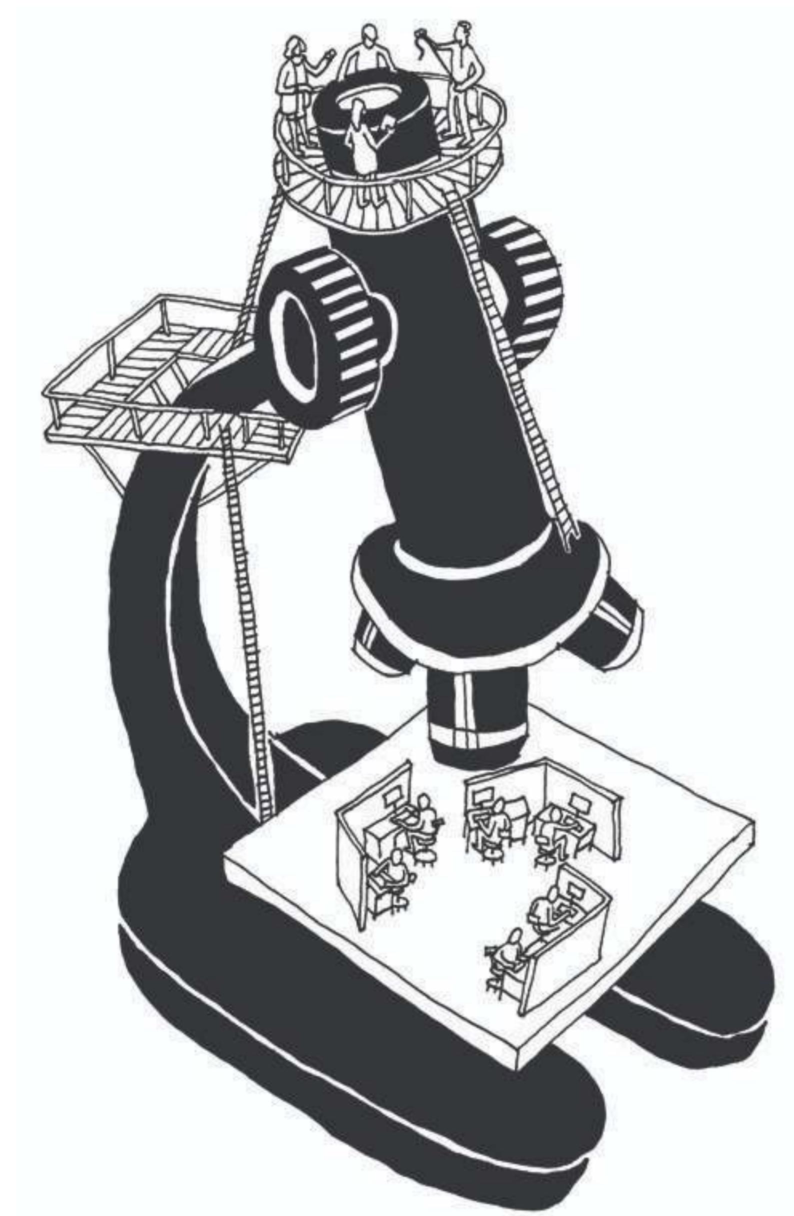
ECS 260: Software Engineering

Lecture 2: Research Methods and Questions

(slides by Eastbrook and Vasilescu)



Planning Checklist

- ✓ Pick a topic
- ❍ Identify the research question(s)
- ❍ Check the literature
- ❍ Identify your philosophical stance
- ❍ Identify appropriate theories
- ❍ Choose the method(s)
- ❍ Design the study
  - Unit of analysis?
  - Target population?
  - Sampling technique?
  - Data collection techniques?
  - Metrics for key variables?
  - Handle confounding factors
- ❍ Critically appraise the design for threats to validity
- ❍ Get IRB approval
  - Informed consent?
  - Benefits outweigh risks?
- ❍ Recruit subjects / field sites
- ❍ Conduct the study
- ❍ Analyze the data
- ❍ Write up the results and publish them
- ❍ Iterate

Choosing a Research Topic

- *Can* it be studied?

vs

- *Should* it be studied?
  - Does it add to the body of knowledge?
  - Who else besides you would care about results?

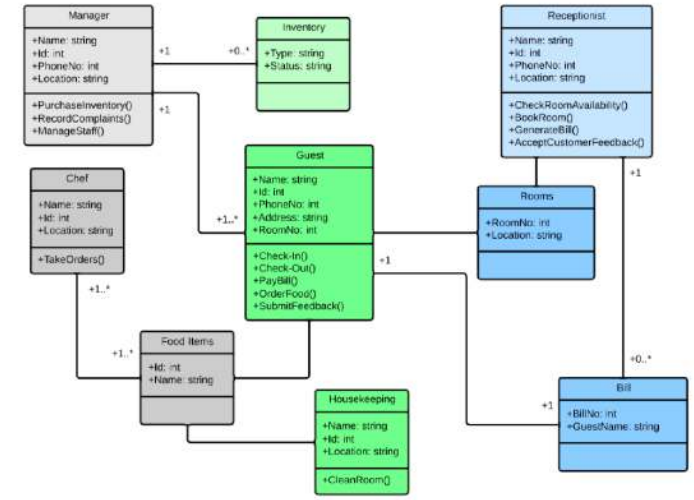
Identify the Research Questions

- Where should the questions come from?
  - Research goals
  - Intuition
  - Social Impact
  - Advisor?
- What makes an RQ good?
  - Precise enough
  - Tests a theory
  - No one else has asked in given context

Scenarios - Jane

- Topic: Effectiveness of a novel fisheye-view file navigator (compact format, less scrolling)
- Intuition: fisheye-view file navigator is more efficient for file navigation
- Critics: the more compact information is difficult to read; developers will not adopt it over the traditional file navigator.
- How can Jane find evidence that supports or refutes her intuition?

Scenarios - Joe



- Topic: understanding how developers in industry use (or not) UML diagrams during software design
- Research goals:
  - Explore how widely UML diagrams are used
  - Explore how UML diagrams are used as collaborative shared artefacts during design
- What strategy can Joe follow?

RQs – First Attempt

- Jane: “Is a fisheye-view file navigator more efficient than the traditional view for file navigation?”
- Joe: “How widely are UML diagrams used as collaborative shared artifacts during design?”

- Thoughts?

RQs – First Attempt

- Both questions are vague, because they make assumptions about:
  - the phenomena to be studied
  - the kinds of situation in which they occur

RECALL!

What type of question are you asking?

- Existence:
  - Does X exist?
- Description & Classification
  - What is X like?
  - What are its properties?
  - How can it be categorized?
  - How can we measure it?
  - What are its components?
- Descriptive-Comparative
  - How does X differ from Y?
- Frequency and Distribution
  - How often does X occur?
  - What is an average amount of X?
- Descriptive-Process
  - How does X normally work?
  - By what process does X happen?
  - What are the steps as X evolves?
- Relationship
  - Are X and Y related?
  - Do occurrences of X correlate with occurrences of Y?
- Causality
  - Does X cause Y?
  - Does X prevent Y?
  - What causes X?
  - What effect does X have on Y?
- Causality-Comparative
  - Does X cause more Y than does Z?
  - Is X better at preventing Y than is Z?
  - Does X cause more Y than does Z under one condition but not others?
- Design
  - What is an effective way to achieve X?
  - How can we improve X?

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- Descriptive-Comparative
  - How does X differ from Y?

Exploratory

- Frequency and Distribution
  - How often does X occur?
  - What is an average amount of X?
- Descriptive-Process
  - How does X normally work?
  - By what process does X happen?
  - What are the steps as X evolves?

Baserate

- Relationship
  - Are X and Y related?
  - Do occurrences of X correlate with occurrences of Y?

Correlation

- Causality
  - Does X cause Y?
  - Does X prevent Y?
  - What causes X?
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Causal Relationship

- Design
  - What is an effective way to achieve X?
  - How can we improve X?

Design

Better RQs
If Little Prior Knowledge

- Existence questions:
  - “Is file navigation something that (certain types of programmers) actually do?”
  - “Is efficiency actually a problem in file navigation?”
  - “Do collaborative shared artifacts actually exist?”
- Description and Classification questions
  - “How can we measure efficiency for file navigation?”
  - “What are all the types of collaborative shared artifacts?”
- Descriptive-Comparative:
  - “How do fisheye views differ from conventional views?”
  - “How do UML diagrams differ from other representations of design information?”

Better RQs

If Some Understanding of the Phenomena

- Frequency and distribution questions
  - “How many distinct UML diagrams are created in large software companies?”
- Descriptive-Process questions
  - “How do programmers navigate files with existing tools?”
- Relationship questions
  - “Does efficiency in file navigation correlate with the programmer’s familiarity with the environment?”
  - “Do managers’ claims about how often they use UML correlate with the actual use of UML?”

- Causality questions:
  - “Do fisheye-views cause an improvement in efficiency for file navigation?”
- Causality-Comparative questions:
  - “Do fisheye-views cause programmers to be more efficient at file navigation than conventional views?”
- Causality-Comparative Interaction questions:
  - “Do fisheye-views cause programmers to be more efficient at file navigation than conventional views when programmers are distracted?”
- Design questions:
  - “What is an effective way for teams to represent design knowledge to improve coordination?”

Literature Review

- Helps you choose a research topic:
  - Determine if the topic is worth studying
  - Limit the scope
- Ask yourself: How does this project contribute to the literature?
  - Addresses a new topic
  - Uses new data collection method
  - Extends the discussion
  - Replicates a study in a new situation
  - Refines / extends a theory

Purpose of Lit Review

- Share with reader results from related studies
- Relate your study to literature, filling in gaps
- Framework for establishing the importance of your study
- Benchmark for comparing results
- **Provide direction for your research questions and hypotheses**
  - E.g., describe the theory that will be used

Forms of Lit Review

- Integrate what others have done and said
- Criticize prior work
- Build bridges between related topics
- Identify the central issues in a field

How to Structure Lit Review

- Hook
- Coherent presentation of what has been done
- The gap, the need for exactly what you did

Easterbrook et al Chapter
- Selecting Methods -

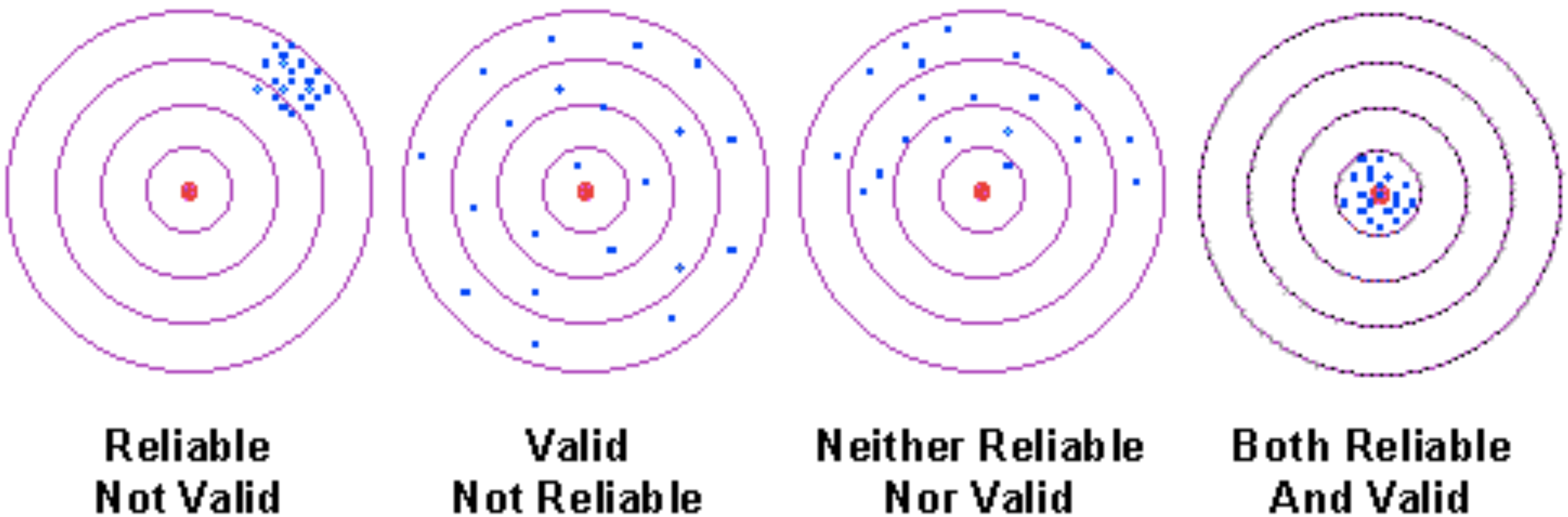
- Controlled Experiments (Quasi-experiments):
  - determine precisely how variables are related
  - or whether a cause–effect relationship exists
- Case Studies (Exploratory/Confirmatory)
  - offer in-depth understanding of how and why certain phenomena occur
- Survey Research
  - identify the characteristics of a broad population

Easterbrook et al Chapter
- Selecting Methods -

- Ethnographies
  - study a community of people to understand how the members of that community make sense of their social interactions
- Action Research
  - attempt to solve a real-world problem while simultaneously studying the experience of solving the problem

Validity vs. Reliability

- Reliability: Does the study get consistent results?
- Validity: Does the study get true results?



Validity (positivist view)

- Construct Validity
  - Are we measuring the construct we intended to measure?
  - Did we translate these constructs correctly into observable measures?
  - Did the metrics we use have suitable discriminatory power?
- Internal Validity
  - Do the results really follow from the data?
  - Have we properly eliminated any confounding variables?
- External Validity
  - Are the findings generalizable beyond the immediate study?
  - Do the results support the claims of generalizability?
- Empirical Reliability
  - If the study was repeated, would we get the same results?
  - Did we eliminate all researcher biases?

Typical Problems

- Construct Validity
  - Using things that are easy to measure instead of the intended concept
  - Wrong scale; insufficient discriminatory power
- Internal Validity
  - Confounding variables: Familiarity and learning;
  - Unmeasured variables: time to complete task, quality of result, etc.
- External Validity
  - Task representativeness: toy problem?
  - Subject representativeness: students for professional developers!
- Theoretical Reliability
  - Researcher bias: subjects know what outcome you prefer

You Gotta Have A Theory

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Meet Stuart Dent

- **Name:**
  - Stuart Dent (a.k.a. "Stu")
- **Advisor:**
  - Prof. Helen Back
- **Topic:**
  - Merging Stakeholder views in Model Driven Development
- **Status:**
  - 2 years into his PhD
  - Has built a tool
  - Needs an evaluation plan

Stu's Evaluation Plan

→ Formal Experiment

- Independent Variable: Stu-Merge vs. Rational Architect
- Dependent Variables: Correctness, Speed, Subjective Assessment
- Task: Merging Class Diagrams from two different stakeholders' models
- Subjects: Grad Students in SE
- H_1: "Stu-Merge produces correct merges more often than RA"
- H_2: "Subjects produce merges faster with Stu-Merge than with RA"
- H_3: "Subjects prefer using Stu-Merge to RA"

→ Results

- H_1 accepted (strong evidence)
- H_2 & H_3 rejected
- Subjects found the tool unintuitive

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What could have gone wrong?

Threats to Validity

- **Construct Validity**
  - What do we mean by a merge? What is correctness?
  - 5-point scale for subjective assessment - insufficient discriminatory power
    - (both tools scored very low)
- **Internal Validity**
  - Confounding variables: Time taken to learn the tool; familiarity
  - Subjects were all familiar with RA, not with Stu-merge
- **External Validity**
  - Task representativeness: class models were of a toy problem
  - Subject representativeness: Grad students as sample of what population?
- **Theoretical Reliability**
  - Researcher bias: subjects knew Stu-merge was Stu's own tool

More on validity in the backup slides at the end of the talk

What went wrong?

- What was the research question?
  - "Is tool A better than tool B?"
- What would count as an answer?
- What use would the answer be?
  - How is it a "contribution to knowledge"?
- How does this evaluation relate to the existing literature?

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Stu's Theory

→ Background Assumptions

- Large team projects, models contributed by many actors
- Models are fragmentary, capture partial views
- Partial views are inconsistent and incomplete most of the time

→ Basic Theory

- (Brief summary:)
- Model merging is an exploratory process, in which the aim is to discover intended relationships between views. 'Goodness' of a merge is a subjective judgment. If an attempted merge doesn't seem 'good', many need to change either of the models, or the way in which they were mapped together.

→ Derived Hypotheses

- Useful merge tools need to represent relationships explicitly
- Useful merge tools need to be complete (work for any models, even if inconsistent)

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Stu's Research Question(s)

Pick just one for now…

- → **Existence**
  - Does model merging ever happen in practice?
- → **Description/Classification**
  - What are the different types of model merging that occur in practice on large scale systems?
- → **Descriptive-Comparative**
  - How does model merging with explicit representation of relationships differ from model merging without such representation?
- → **Causality**
  - Does an explicit representation of the relationship between models cause developers to explore different ways of merging models?
- → **Causality-Comparative**
  - Does the algebraic representation of relationships in Stu's tool lead developers to explore more than do pointcuts in AOM?

Stu's Method(s) Selection…

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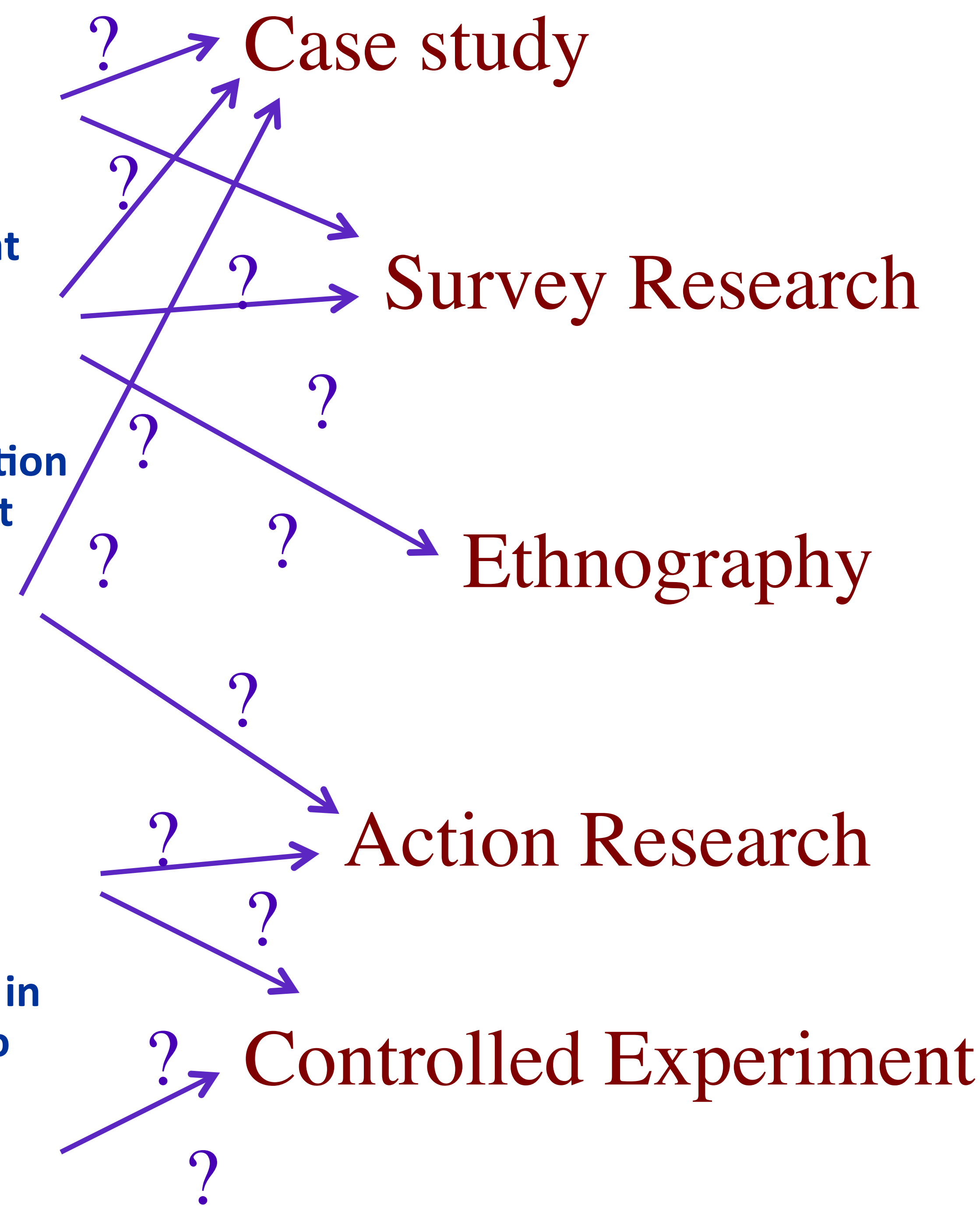

Warning

No method is perfect

Don't get hung up on methodological purity

Pick something and get on with it

Some knowledge is better than none

Why Build a Tool?

- Build a Tool to Test a Theory
  - Tool is part of the experimental materials needed to conduct your study
- Build a Tool to Develop a Theory
  - Theory emerges as you explore the tool
- Build a Tool to Explain your Theory
  - Theory as a concrete instantiation of (some aspect of) the theory

Why did we build tools?

Why did Stu build a tool?

?

Why do you build tools?